High-Temperature Electrical Conductivity Measurement Device for Composite Wires

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Final Report

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Chapter 1

1.1 Abstract

Wires are a standard for carrying electrical charges across distances for conductive systems. To monitor the performance of a wire, its electrical conductivity is measured using a four-point probe design. To further measure conductive properties, the design of a reusable 4-point probe that can withstand high temperatures while measuring changes in resistance was developed to fit within an existing furnace setup. The four-point probe design that was used was made to withstand 260℃, with the use of Slippery Sheet made from Teflon® PTFE as the main build material, used because of its high temperature range up to 260℃, machineability, and insulating properties. The fabrication of a working prototype was made to have a clamping mechanism with the use of torsion springs coupled to the side and stabilized by a an m6 long machine bolt to pin down the four wires that come into contact with the sample.

1.2 Research Goals

There was already a prototype designed to measure the electrical conductivity of a wire sample at a high temperature, but it was fabricated as a model and was not able to be used over consecutive trials. The prototype used electrical tape wrapped around the base to secure gold wires in a four-point probe arrangement, ensuring enough pressure is applied such that the wires remain still and pinned to the sample. With this, a conductivity test could be completed, but upon starting another test on a new sample, the electrical tape would need to be removed and reapplied, creating unnecessary uncertainty in the procedure. The goal of this project is to refine the existing model with one that can execute the same functions while using a more sustainable design. In this case, a sustainable device would be able to test multiple samples without having to completely deconstruct the apparatus between trials. The new device must fulfill several

requirements relating to size and conductivity. Since tests will be conducted in a tube furnace with a diameter of one inch, the device must be small enough to fit into that space. Anything too large would be considered unsustainable. Like the electrical tape, the device itself must be able to provide sufficient pressure to the four wires so that they remain in place and separated from each other. This is especially critical as low pressures will skew data and possibly lead to an electrical short. To ensure a steady, consistent flow of electricity, the device must be made of nonconducting material. Therefore, common metals such as steel and aluminum cannot be used, despite their great machinability and malleability properties. Additionally, all materials used in the device must be able to withstand 260℃, as samples will be tested at that temperature. Given these constraints, the device must be used to measure the electrical conductivity of a sample at 260℃. The project will be considered a success if this can occur and if our model is built in such a way that another test could happen again.

Chapter 2

2.1 Literature Review

Comprehensive reviews of the four-point probe concept demonstrate several key concepts that allow for the measurement of electrical conductivity. Schuetze et al. compiled several applications of this technique to aluminum test structures and semiconductors. Here, four electrodes are placed in a line, equidistant from one another, with the two outer electrodes supplying the current and the two inner ones measuring resistivity, as seen in Figure 1.

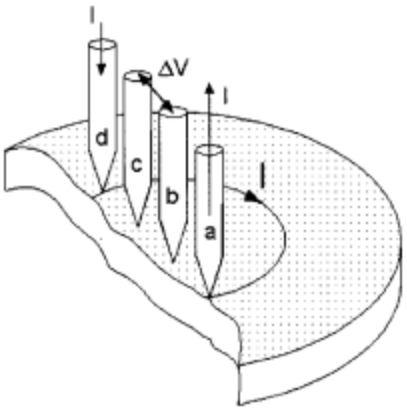


Figure 1 - Four-point probe setup along the edge of a material (Schuetze et al.)

Though Schuetze et al. are concerned more with large volumes of material, their technique can be applied to composite wires as well.

Different materials were considered when deciding how to create the base. The base had to be easily machinable, have good thermal and electrical insulation properties, and be able to withstand temperatures up to 260°C. Tychanicz-Kwiecień et al. provided listings of several heat-resistant and nonconducting materials. Ceramic and glass fibers could withstand 300°C, basalt fibers were 700°C, and aluminosilicate and zirconium were 1000°C and 1600°C, respectively. Though these materials could withstand high temperatures, they each had critical problems. Ceramic and fibrous materials are difficult to machine on a conventional mill, as outlined by Bharathi et al., and pose a significant challenge to creating precise cuts for the base. These materials can also be toxic at high temperatures, releasing harmful nanoparticles according to Bessa et al. With these concerns in mind, ceramics and fibers were discarded in favor of Teflon PTFE. Blumm et al. highlighted the advantages of PTFE in its insulation, machinability, and safety. At temperatures above 100°C, PTFE materials have a thermal conductivity of 0.25 $\frac{W}{m \cdot K}$, which is low enough to reduce any significant change in material properties.

2.2 Methods

As project development began, two designs emerged that utilized spring force to provide enough downward pressure on the wires while keeping them separated from each other. Springs were chosen as the force to be exerted because they are easily scalable to accommodate for size constraints, and reusable because of their restoring force. Both designs were made using SolidWorks CAD. Design 1 featured vertical extension springs alongside the base (Figure 2).

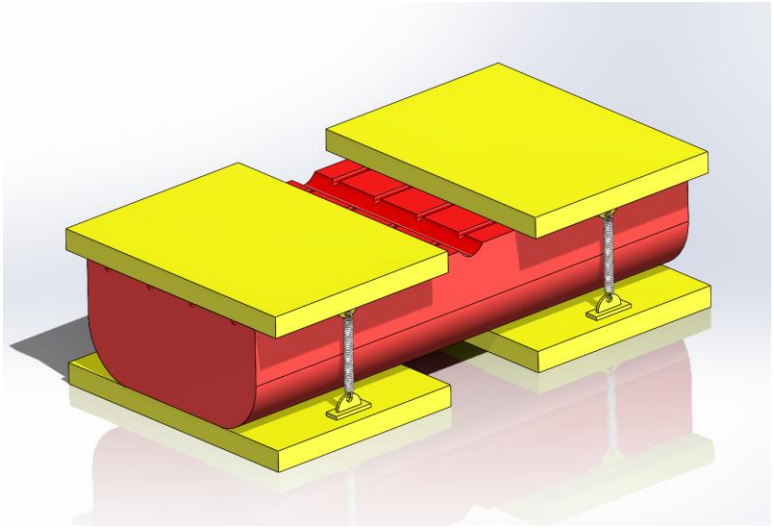


Figure 2 - Vertical extension springs providing pressure to the wires

The model consists of four 0.25 in. extension springs that have a spring rate of 11.1 lbs./in. When fully stretched, the spring would have a maximum load of 1.09 lbs. that could effectively support an even pressure on all four wires. The model also leaves plenty of room for the wire sample in the center, allowing for simply exchanges between trials. Though it appeared practical, there were some glaring challenges regarding its production. The small springs were the primary issue as they required a connection to small holes that were extremely difficult to properly machine and attach. Given the small size and high temperatures, the connection points were predicted to be a weak point in the model that would certainly succumb to intense heat. The extension springs may be able to apply uniform pressure, but they would not be able to do that

over a long period of time. Design 2 featured horizontal torsion springs placed inside of the base to conserve space (Figure 3).

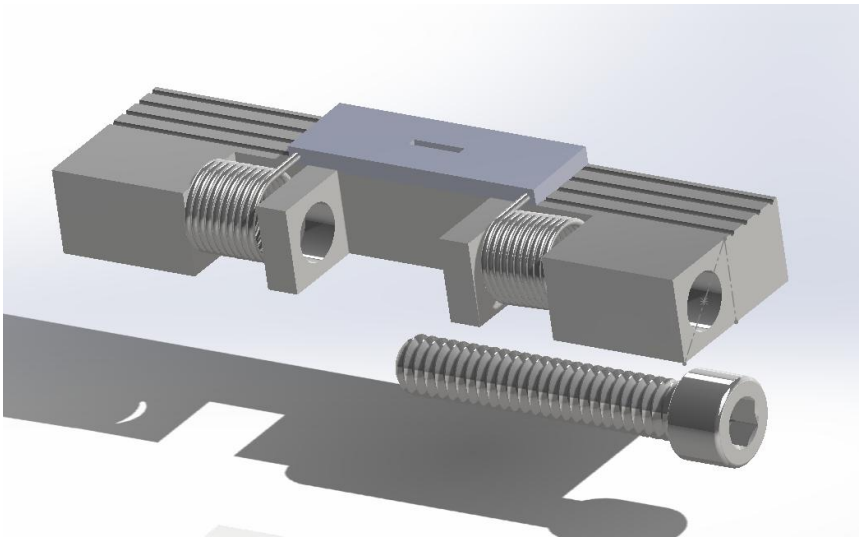


Figure 3 - Horizontal torsion springs, supported by a bolt, providing pressure to the wires

A singular top lid replaced the four exterior pieces used in the first design with a single piece containing a slit. Two small holes would be drilled into the top piece so that it could be supported by the springs without exposing the wires to the metal conducting wires of the springs. Though the slit was smaller than the opening in the first design, it would still allow a sample to oxidize and exert more downward pressure to wires as they approached the sample over a smaller area. The bottom wires of the springs and the wires inside of the upper piece would be glued using a nonconducting epoxy solution. To support the springs and keep them on the same plane, a bolt was drilled through the base. The m6 bolt was narrow enough to maintain the integrity of the base structure, but thick enough to effectively support the springs. Small grooves would be etched into the base to keep the wires in place. The second design, with its reliability and easier assembly, was chosen over the first design.

The manufacturing process began with a $3 \times 2 \times 0.75\ in^3$ piece of Teflon PTFE. A bandsaw was used to cut the block into six strips that could act as the base. From there, the

flattest two pieces were reserved while the other four were cut even smaller to be the upper piece. 1 mm holes were drilled into the upper piece as it was secured with a vice clamp to maintain stability and ensure a consistent hole. For the base, a drill press was used to create a hole that would hold the bolt. Then, a milling machine was used to cut out holes in the base for the torsion springs. Finally, a mixture of epoxy resin and epoxy hardener was placed inside of the holes of the top piece and along the bottom of the base to hold the extension springs in place. After waiting for the epoxy to cure, the device assembly was complete (Figure 4).

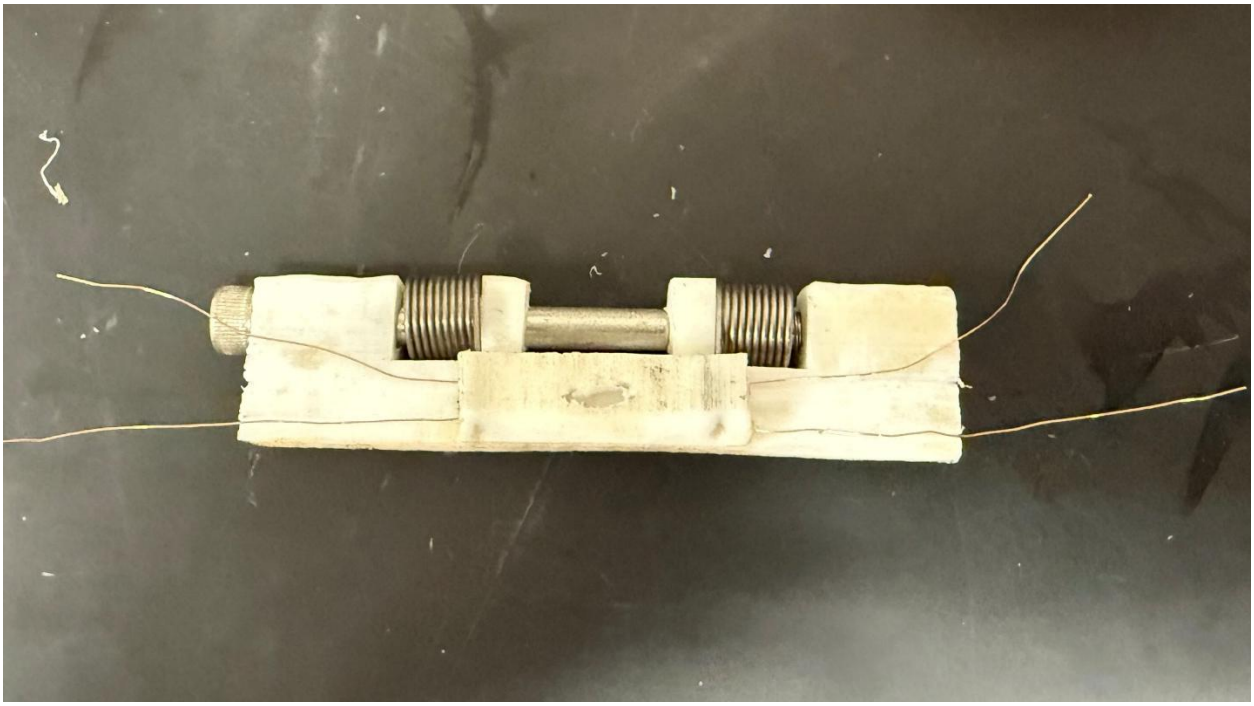


Figure 4 - The completed assembly

Chapter 3

3.1 Testing

The assembled device experienced both a thermal and conductivity test. For the thermal test, a thermocouple was taped underneath the device and placed inside of the tube furnace. As the tube furnace heated, the thermocouple measured the temperature inside of the tube relative to the reading that the tube furnace gave. This ensured that the temperature readings from the

thermocouple were consistent with the actual readings given by the tube furnace. The tube furnace was heated to 260℃, then observed. After about ten minutes, the device began to smoke from inside the tube. Upon removing it, it was apparent that excess epoxy had spread along the top and sides of the base and began burning in the tube.

As for the conductivity test, four wires were placed on the base in an outer-inner opposing pattern. A known current was applied to the circuit through the primary outer wire, propagating through the sample and into two intermediate sensing wires located between the outer conductors. Resistivity is determined by measuring the differential current velocities at the sensing points, with the current exiting the circuit via the secondary outer wire, as depicted in Figure 5.

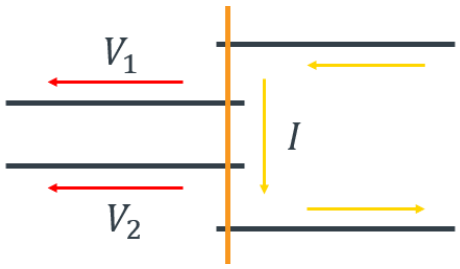
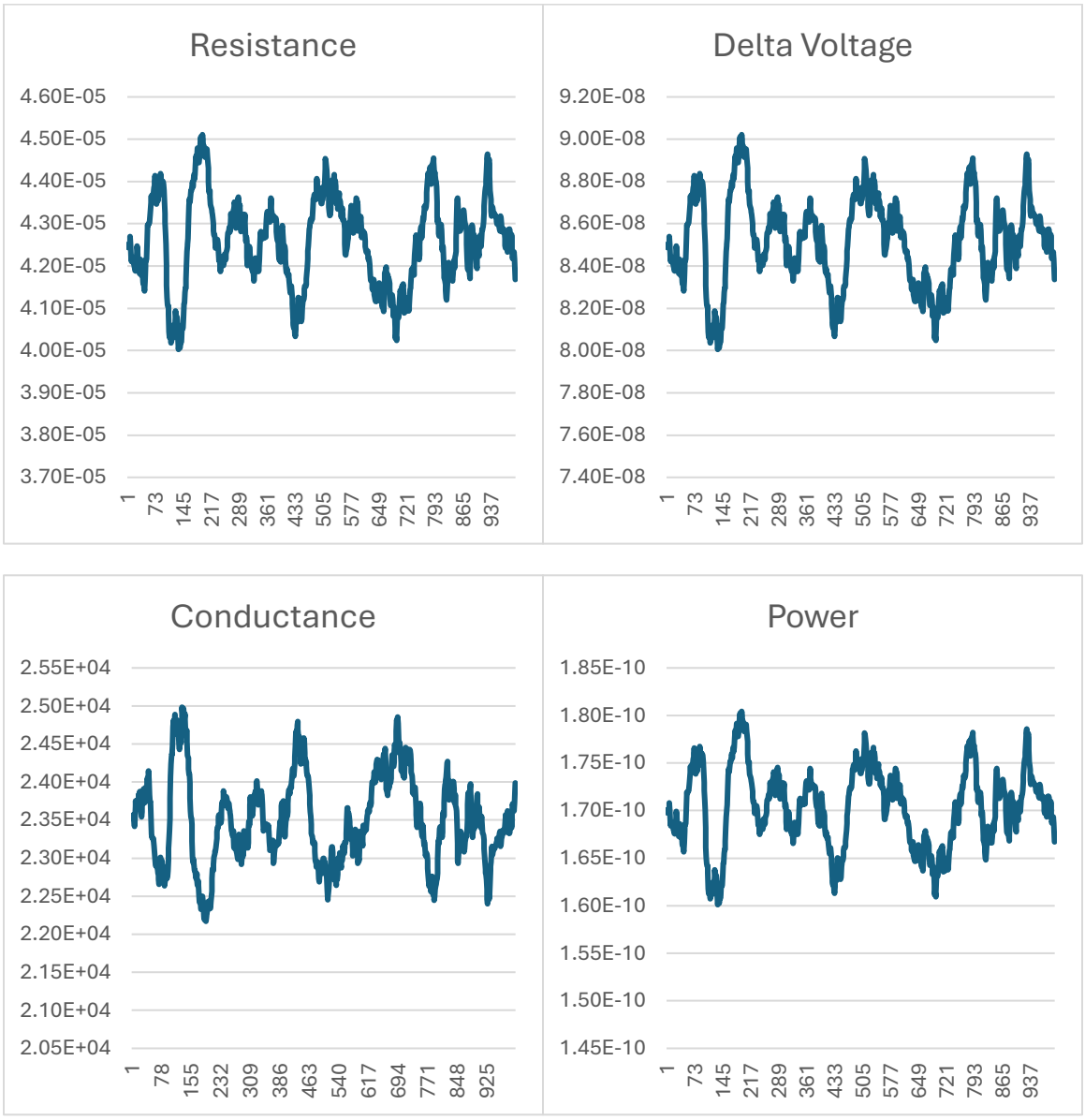


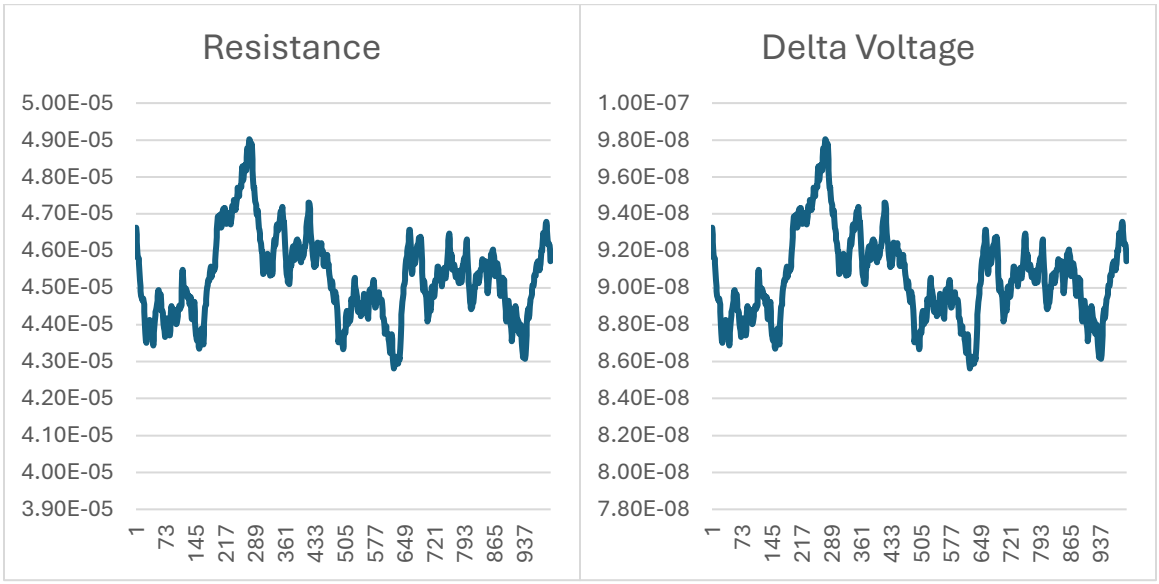
Figure 5 – Four-Point Probe layout; $R = \frac{\Delta V}{I}$

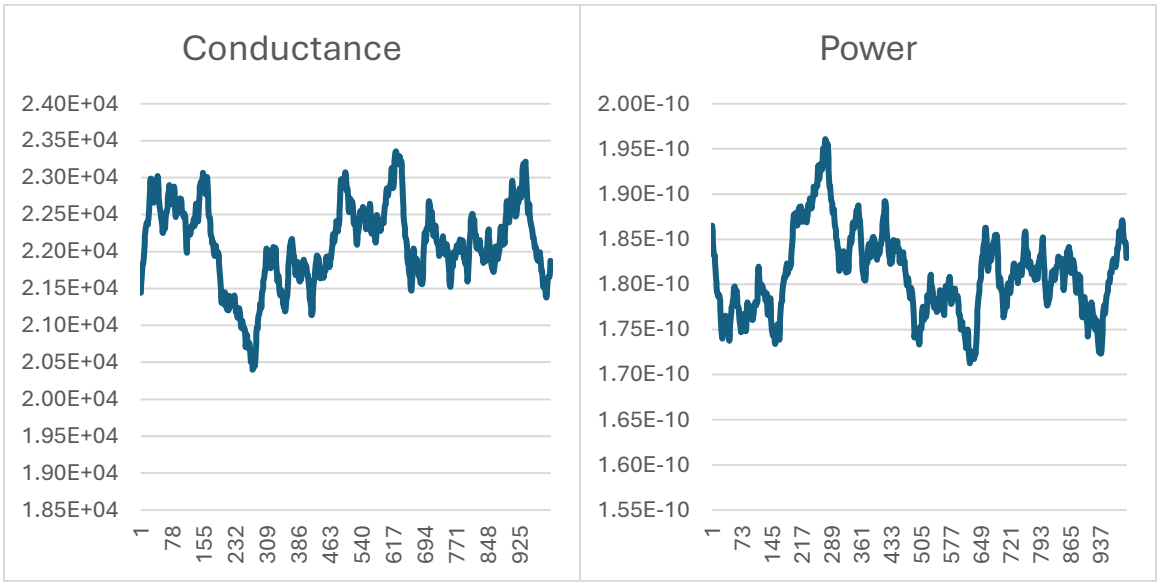
To ensure there was a proper connection between the outer wires, the sample, and the inner sensing wires, a multimeter was used to check that all possible connection combinations displayed a voltage value. Four tests were run in total, with one test to serve as the control to the data. Data was recorded using software that measures resistivity, delta voltage, conductivity, and power values recorded over 1000 data points respectively.

Results of Applied Pressure Test 1:

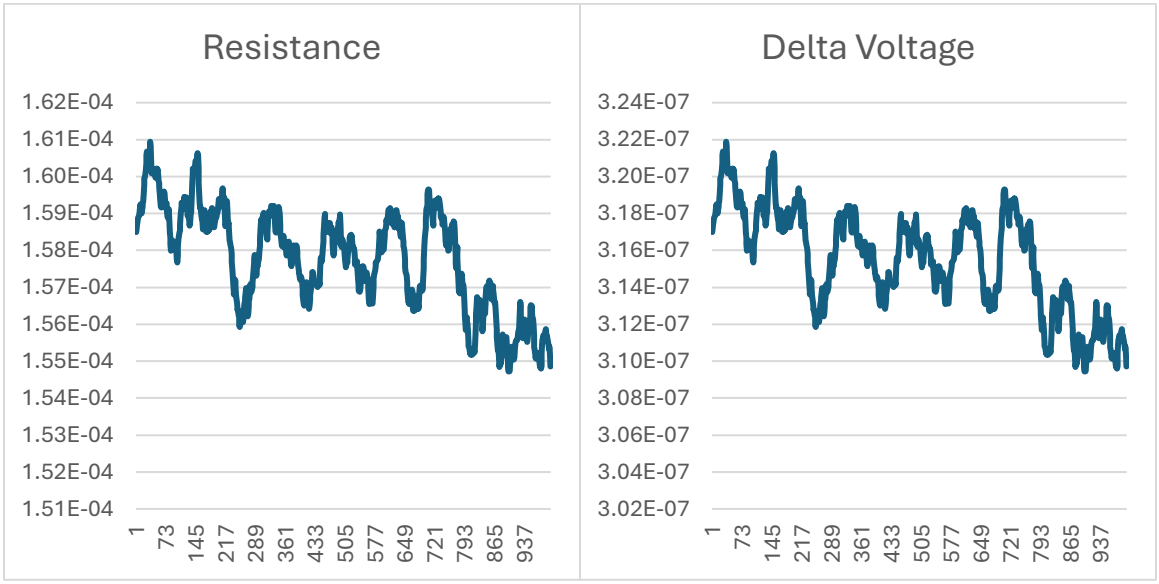


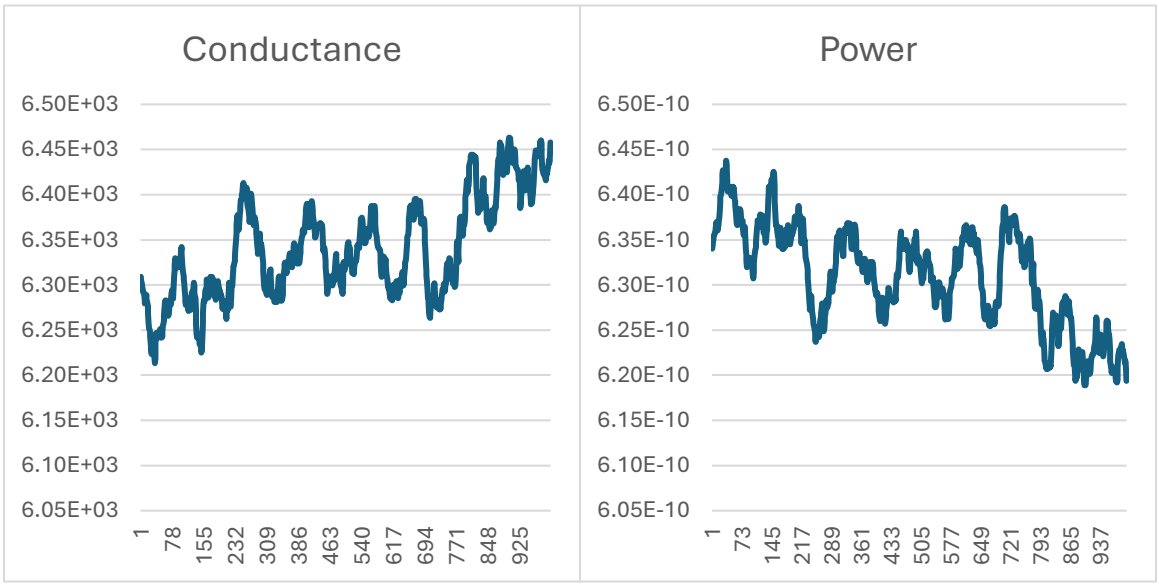
Results of Applied Pressure Test 2:



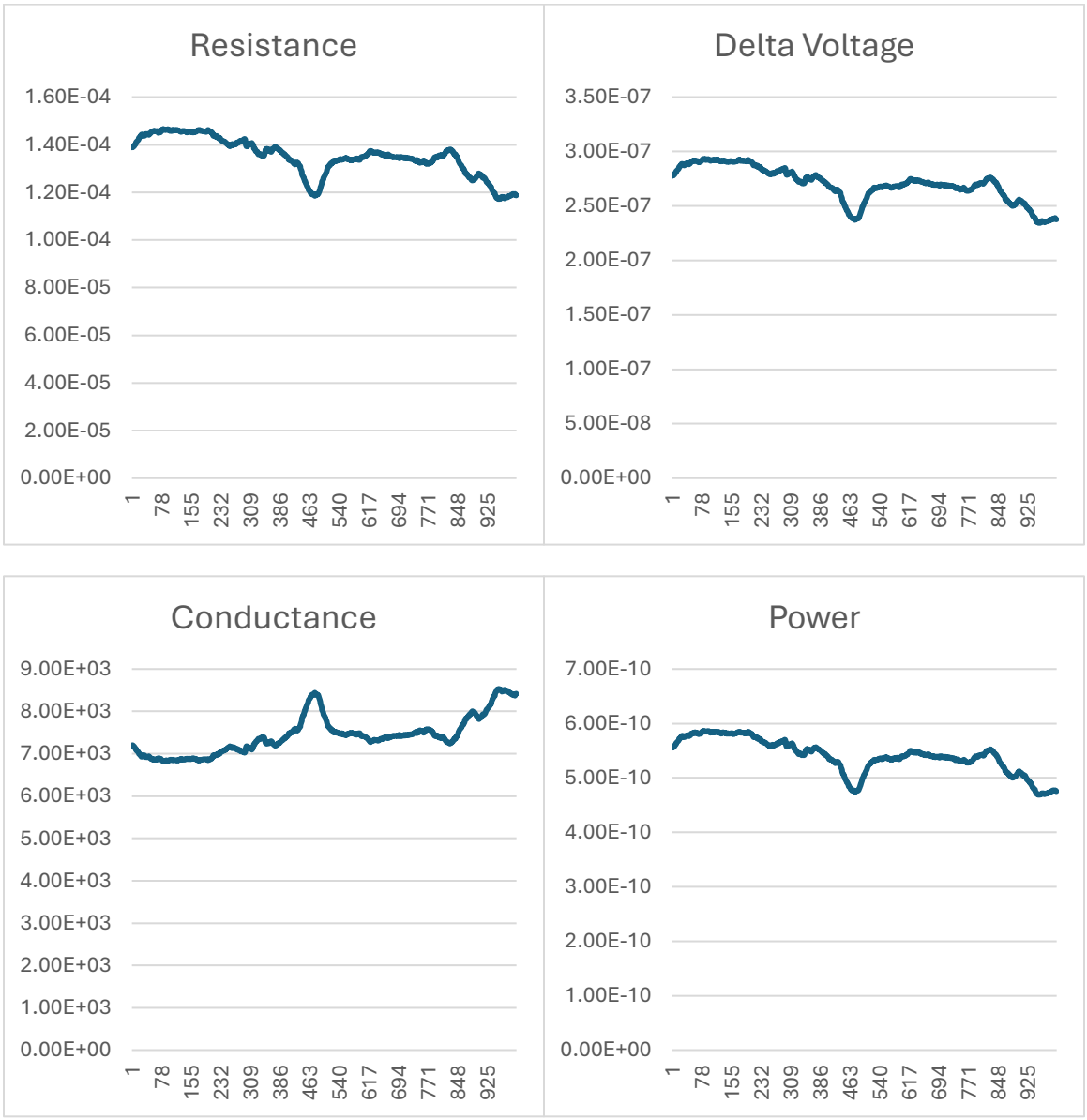


Results of No Applied Pressure Test 1





Results of No Applied Pressure Test 2:



3.2 Discussion/Analysis

During the thermal test, it is believed that the epoxy mixture caused the device to smoke while in the tube furnace. This may have occurred due to improper mixing or incorrect application. The mixture called for a 5:1 ratio of epoxy resin to epoxy hardener, which was mixed in a beaker. It is possible that there were errors in stirring and in estimating the mixture ratio. In addition, thin layers of the liquid mixture may have spread to other parts of the base when drying, leaving them

vulnerable to high temperatures. Besides the epoxy, the rest of the model was able to withstand the heat, as predicted by manufacturers' descriptions.

After performing the four-point probe connectivity tests, four key parameters were recorded: resistance, delta voltage, conductance, and power. The data collected under conditions of no additional applied pressure on the probe lid demonstrated consistent proportional trends among resistance, voltage, and power, while conductance exhibited the expected inverse behavior relative to resistance.

Notably, fluctuations were observed in the minimum and maximum values across all parameters, indicating variability in the sample or contact stability during testing. These fluctuations suggest potential sources of measurement uncertainty, particularly in resistance and voltage readings, which subsequently affect calculated conductance and power values.

Regarding the four-point probe connectivity tests under conditions of additional applied pressure on the probe lid, there was no change in the relationship between resistance, voltage, conductivity, and power. However, the trend of the graphs is more evident and demonstrates an observable linear increase over the points in time, meaning a better contact on the sample by the wires likely reduced potential sources of measurement uncertainty.

Chapter 4

4.1 Future Work

Though the current design was able to capture results, several improvements can still be made. The base material, currently, Teflon PTFE, could be exchanged for a less elastic material that would allow the wires to better conform to their individual slits under high temperatures. It is also possible to manufacture the base better using more technical and precise machinery. For this project, a bandsaw, drill press, and mill were used to make cuts, however, a CNC machine would

be able to execute commands with much more consistency. Since the lack of space was a key constraint, CNC could account for small tolerances, allowing for more complex concepts. A more advanced clamping mechanism could also be worked out. Instead of using springs, the base and an epoxy mixture could be applied to wires that fit inside of the base, removing the need for an external piece that would just take up more space. Though our model was successful in running several tests, more precise measurement and manufacturing devices could improve its effectiveness and reliability.

Appendix

A. References

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